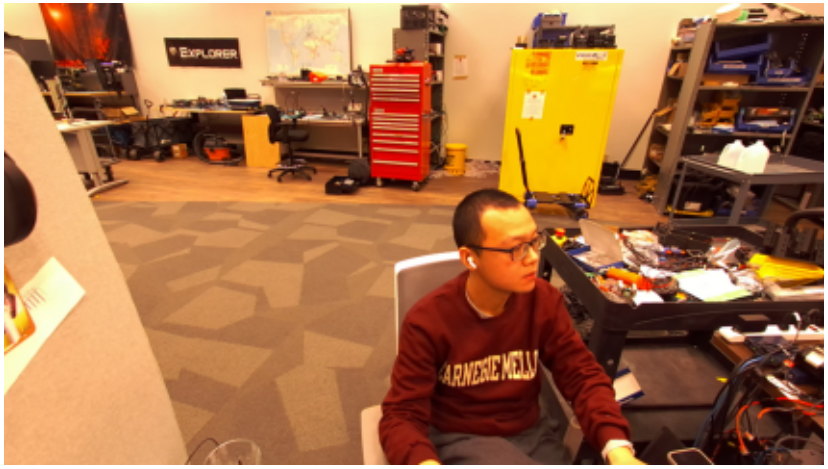
ZED depth: ONNX vs PyTorch

PASS: outputs within tolerance

Model: neural_depth_5.3.onnx | atol=0.003, rtol=0.003

Left RGB input: im0.png



Right RGB input: im1.png



Output	Mean abs.	Max abs.	RMSE	P99 abs.	Outside tol.	Result
disparity	2.27053e-06	0.000132561	5.76625e-06	2.28882e-05	0%	PASS
disparity_right	0	0	0	0	0%	PASS
confidence	4.57436e-07	5.73993e-05	1.73036e-06	7.15747e-06	0%	PASS

Input HxW: [621, 1104] | network HxW: [256, 512] | bottom/right padding: [0, 0]

Output shape: [1, 1, 128, 256] | comparison includes padding; disparity is not metric depth.

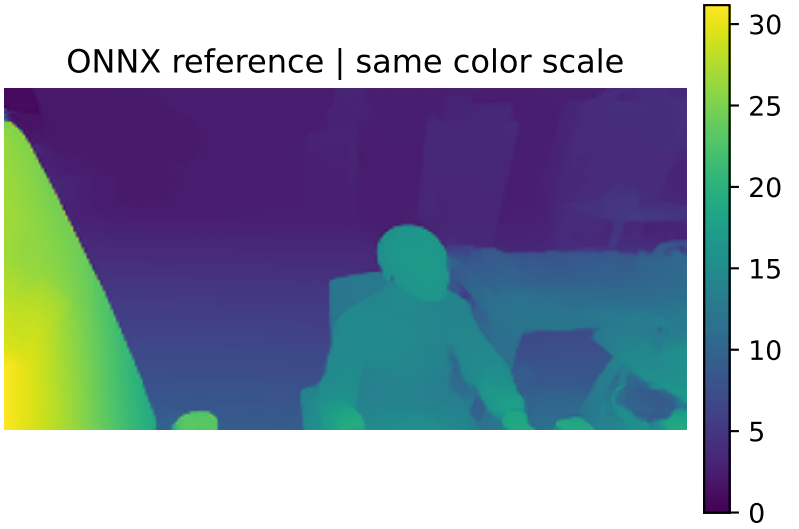
Reference: ONNX Runtime standard layers + independent NumPy custom operators. No PyTorch tensors are injected.

disparity_right is a zero-filled placeholder; matching zeros does not validate right-view depth.

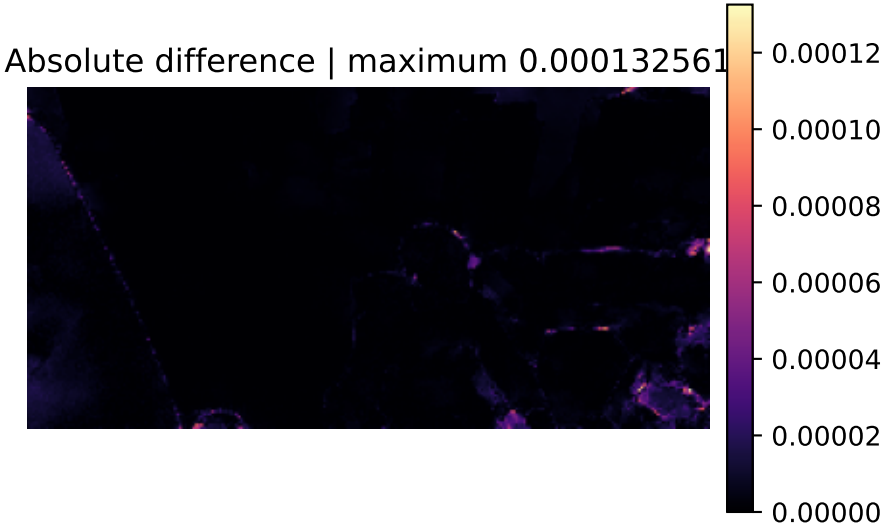
This measures implementation agreement on one stereo pair. No ground truth or native ZED CUDA parity was tested.

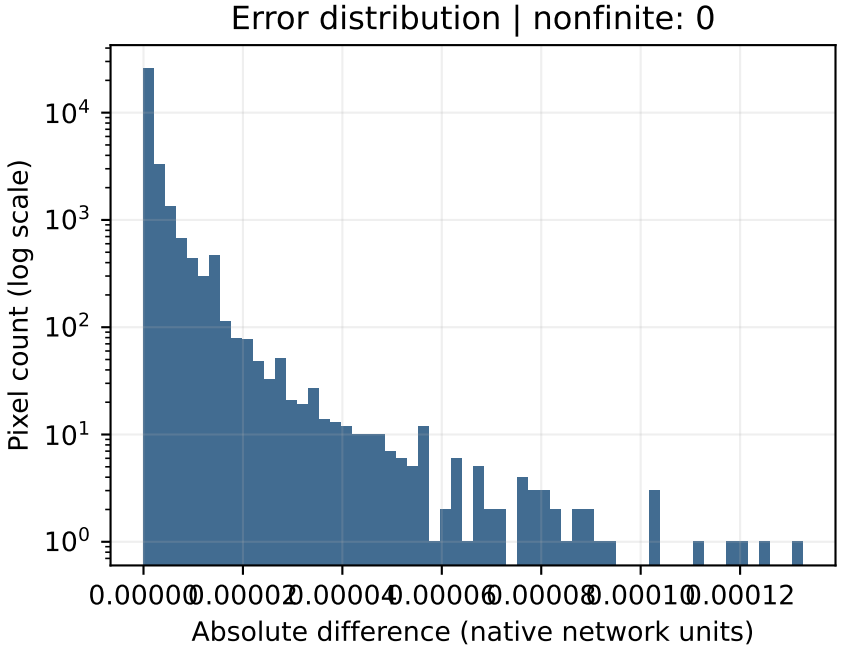
Timing: PyTorch forward 0.146s; reference setup + run 0.314s.

Disparity output analysis



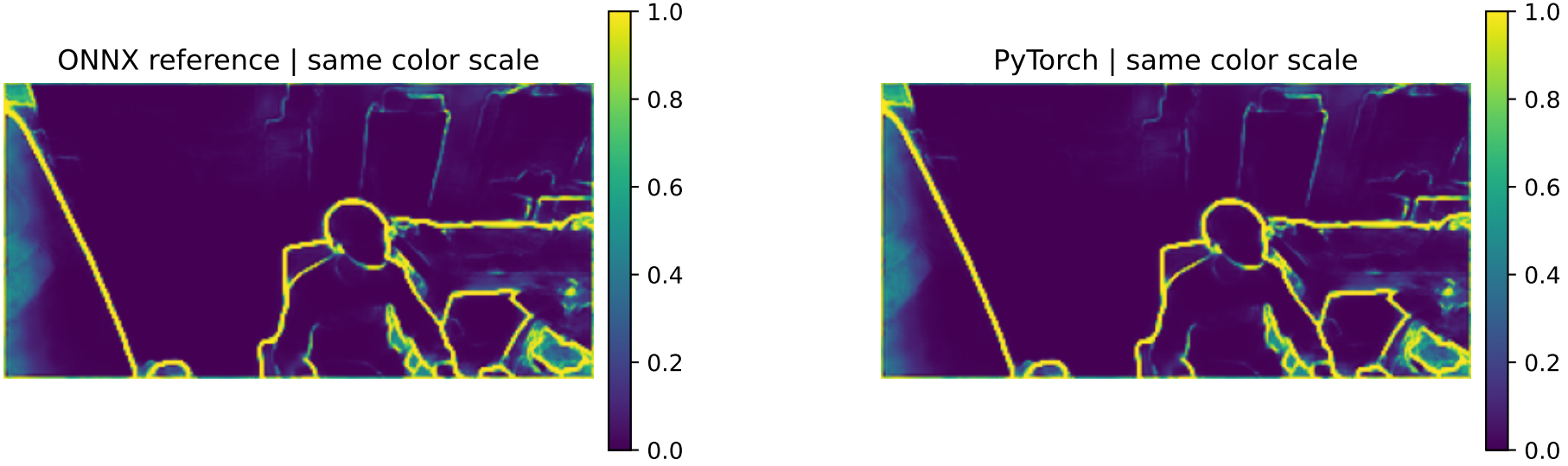


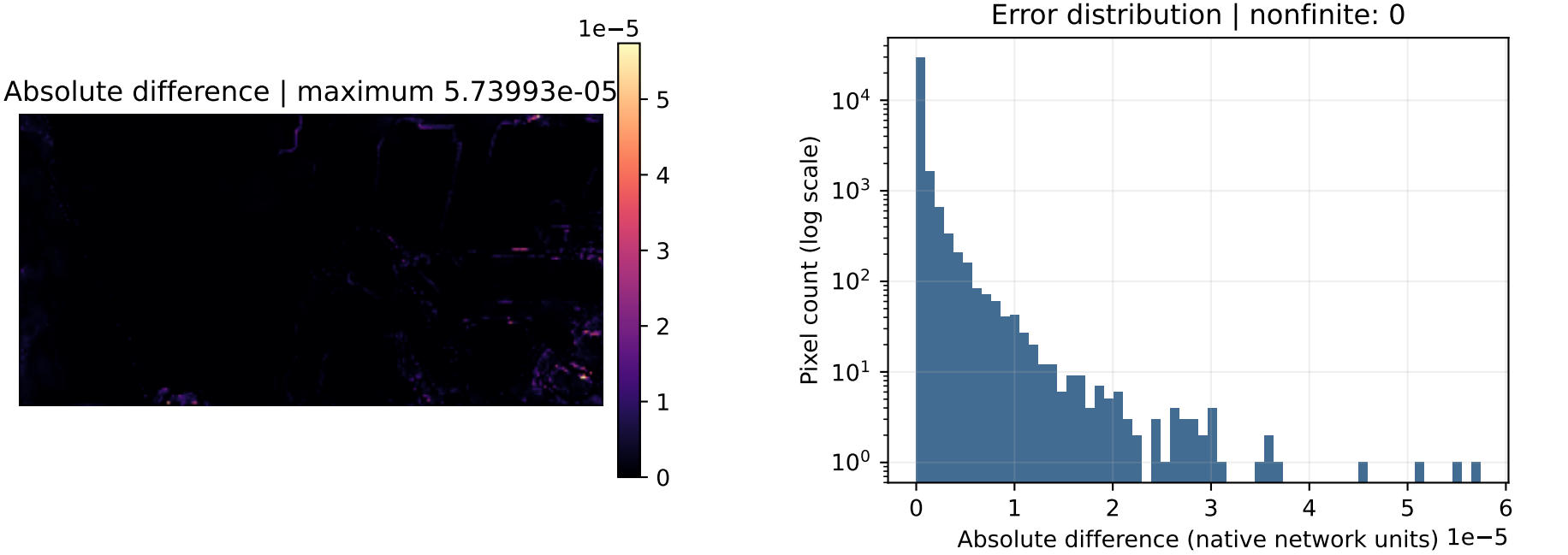




Out-of-tolerance: 0% | Difference colors are magnified to the maximum error; inspect numeric values, not color alone.

Confidence output analysis





Out-of-tolerance: 0% | Difference colors are magnified to the maximum error; inspect numeric values, not color alone.